Answers: Standard Normal Areas Practice

Find each area under the standard normal curve. Use the standard normal table, or a spreadsheet with `NORMSDIST`. The same problems are in this spreadsheet: NormalAreasPractice-Answers.xlsx.

Area	Spreadsheet formula
Left tail area	`=NORMSDIST(z)`
Right tail area	`=1.0-NORMSDIST(z)`
Area between z_1 and z_2	`=NORMSDIST(z2)-NORMSDIST(z1)`

Left Tail Areas

#	z	Left Tail Area	Formula
1	-1.33	0.0918	`=NORMSDIST(-1.33)`
2	0.52	0.6985	`=NORMSDIST(0.52)`
3	-0.33	0.3707	`=NORMSDIST(-0.33)`
4	1.86	0.9686	`=NORMSDIST(1.86)`
5	-2.15	0.0158	`=NORMSDIST(-2.15)`
6	0.08	0.5319	`=NORMSDIST(0.08)`
7	-0.91	0.1814	`=NORMSDIST(-0.91)`
8	2.34	0.9904	`=NORMSDIST(2.34)`
9	-1.62	0.0526	`=NORMSDIST(-1.62)`
10	1.25	0.8944	`=NORMSDIST(1.25)`

Right Tail Areas

#	z	Right Tail Area	Formula
1	1.42	0.0778	`=1.0-NORMSDIST(1.42)`
2	-0.64	0.7389	`=1.0-NORMSDIST(-0.64)`
3	2.05	0.0202	`=1.0-NORMSDIST(2.05)`
4	-1.38	0.9162	`=1.0-NORMSDIST(-1.38)`
5	0.47	0.3192	`=1.0-NORMSDIST(0.47)`
6	-2.21	0.9864	`=1.0-NORMSDIST(-2.21)`
7	1.73	0.0418	`=1.0-NORMSDIST(1.73)`
8	-0.15	0.5596	`=1.0-NORMSDIST(-0.15)`
9	0.96	0.1685	`=1.0-NORMSDIST(0.96)`
10	-1.09	0.8621	`=1.0-NORMSDIST(-1.09)`

Areas Between

#	z_1	z_2	Area Between	Formula
1	-0.94	1.27	0.7243	`=NORMSDIST(1.27)-NORMSDIST(-0.94)`
2	-1.96	1.96	0.9500	`=NORMSDIST(1.96)-NORMSDIST(-1.96)`
3	0.25	1.75	0.3612	`=NORMSDIST(1.75)-NORMSDIST(0.25)`
4	-2.30	-0.40	0.3339	`=NORMSDIST(-0.40)-NORMSDIST(-2.30)`
5	-1.15	0.65	0.6171	`=NORMSDIST(0.65)-NORMSDIST(-1.15)`
6	0.60	2.45	0.2671	`=NORMSDIST(2.45)-NORMSDIST(0.60)`
7	-0.72	0.38	0.4123	`=NORMSDIST(0.38)-NORMSDIST(-0.72)`
8	-1.80	-1.05	0.1109	`=NORMSDIST(-1.05)-NORMSDIST(-1.80)`

#	z_1	z_2	Area Between	Formula
9	1.12	2.60	0.1267	`=NORMSDIST(2.60)-NORMSDIST(1.12)`
10	-2.58	1.44	0.9201	`=NORMSDIST(1.44)-NORMSDIST(-2.58)`

Answers come from the spreadsheet. With the printed table you may get a last digit that differs by 1 (for example 0.7244 for #1 of the areas between).